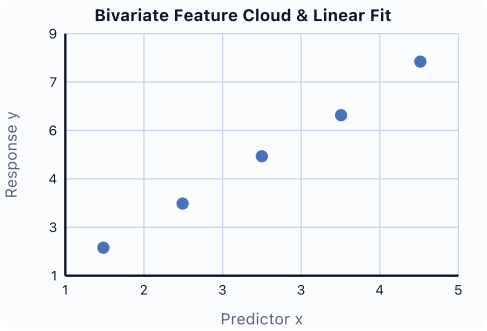
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

PCA & Dimensionality Reduction Practice Exercises

1 MCQ

Principal Component Analysis (PCA) projects a dataset from R^{50} onto its top 6 principal components. What is the dimension of each transformed data point?



A 25

B 50

C 44

D 6

2 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?

A 62.0%

B 86.0%

C 24.0%

D 72.1%

3 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 64$, $\lambda_2 = 23$, and $\lambda_3 = 13$. What percentage of total variance is explained by the first principal component PC_1?

A 64.0%

B 87.0%

C 23.0%

D 73.6%

4 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1?

A 66.0%

B 88.0%

C 22.0%

D 75.0%

5 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

A 68.0%

B 89.0%

C 21.0%

D 76.4%

6 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1?

A 70.0%

B 90.0%

C 20.0%

D 77.8%

7 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1?

A 72.0%

B 91.0%

C 19.0%

D 79.1%

8 MCQ

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 74$, $\lambda_2 = 18$, and $\lambda_3 = 8$. What percentage of total variance is explained by the first principal component PC_1?

A 74.0%

B 92.0%

C 18.0%

D 80.4%

9 MCQ

Consider a 2D data point $\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$. What is the squared Euclidean norm (or squared magnitude) of this vector?

A 7

B 9

C 16

D 25

10 MCQ

A binary classifier $p(t)$ produces the confusion matrix below with $\mathrm{TP} = 68$, $\mathrm{FP} = 10$, $\mathrm{FN} = 48$, and $\mathrm{TN} = 80$. Determine the total number of correct classifications $q = \mathrm{TP} + \mathrm{TN}$.

A 143

B 148

C 165

D 152

11 MCQ

Consider a dataset with two features, X_1 and X_2. The covariance matrix for this dataset is given by:

$$\Sigma = \begin{pmatrix} 10 & 5 \\ 5 & 20 \end{pmatrix}$$

What is the variance of the first feature, X_1?

A 10

B 5

C 20

D 25

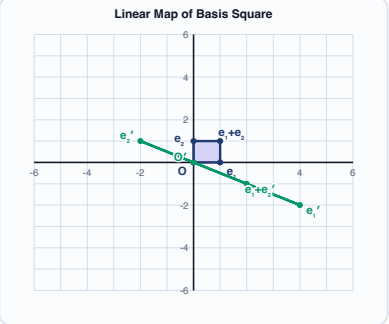
12 MCQ

Consider a dataset of observations $\{x_i\}_{i=1}^{n}$ where each x_i is a 2-dimensional vector. The sample covariance matrix for this dataset is given by:

$$S = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}$$

Principal Component Analysis (PCA) is applied to this dataset. The first principal component corresponds to the direction with the largest variance. Let the direction vector of the first principal component be $\mathbf{v}_1 = \begin{pmatrix} a \\ b \end{pmatrix}$.

The variance explained by this principal component is given by the corresponding eigenvalue λ_1.
If $\mathbf{v}_1$ is a unit vector, what is the value of $a^2 + b^2$?



A 2

B 4

C 1

D 5

13 MCQ

Consider a dataset of n data points, each residing in R^D. We are performing Principal Component Analysis (PCA) to reduce the dimensionality. The first three principal components capture 70% of the total variance, the first six capture 85%, and the first ten capture 92%. You are tasked with selecting a subset of principal components for a downstream machine learning model. If the model requires that at least 90% of the original variance be retained, and you want to use the minimum possible number of principal components, what is the dimensionality of the transformed data?

A D

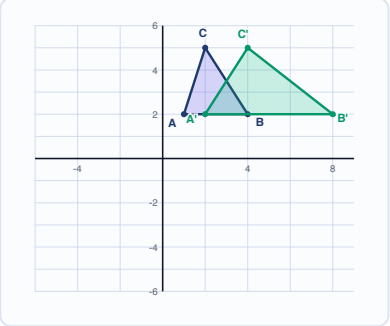
B 3

C 6

D 10

14 MCQ

Consider a 2D dataset represented by the points $A = (1, 2)$, $B = (4, 2)$, and $C = (2, 5)$. If this dataset undergoes a linear transformation represented by the matrix $M = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, what are the coordinates of the transformed points A', B', and C'?



A $A' = (2, 2), B' = (8, 2), C' = (4, 5)$

B $A' = (2, 4), B' = (8, 4), C' = (4, 10)$

C $A' = (3, 2), B' = (6, 2), C' = (4, 7)$

D $A' = (1, 4), B' = (4, 4), C' = (2, 10)$

15 MCQ

In an applied problem, a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 66$, $\lambda_2 = 22$, and $\lambda_3 = 12$. What percentage of total variance is explained by the first principal component PC_1?

A 22.0%

B 75.0%

C 66.0%

D 88.0%

16 MCQ

Suppose a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 68$, $\lambda_2 = 21$, and $\lambda_3 = 11$. What percentage of total variance is explained by the first principal component PC_1?

A 89.0%

B 68.0%

C 21.0%

D 76.4%

17 MCQ

An observed 3D dataset has covariance matrix eigenvalues $\lambda_1 = 70$, $\lambda_2 = 20$, and $\lambda_3 = 10$. What percentage of total variance is explained by the first principal component PC_1?

A 90.0%

B 20.0%

C 77.8%

D 70.0%

18 MCQ

Consider a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 72$, $\lambda_2 = 19$, and $\lambda_3 = 9$. What percentage of total variance is explained by the first principal component PC_1?

A 91.0%

B 19.0%

C 72.0%

D 79.1%

19 MCQ

In an applied problem, a 3D dataset has covariance matrix eigenvalues $\lambda_1 = 76$, $\lambda_2 = 17$, and $\lambda_3 = 7$. What percentage of total variance is explained by the first principal component PC_1?

A 76.0%

B 81.7%

C 93.0%

D 17.0%

20 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 14$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times 14$ rectangular matrix

B $N \times N$ symmetric matrix

C 14×14 skew-symmetric matrix

D 14×14 symmetric matrix

21 MCQ

A centered feature matrix $\mathbf{X}$ has shape $N \times 18$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A 18×18 symmetric matrix

B $N \times N$ symmetric matrix

C 18×18 skew-symmetric matrix

D $N \times 18$ rectangular matrix

22 MCQ

In an applied problem, a centered feature matrix $\mathbf{X}$ has shape $N \times 22$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times N$ symmetric matrix

B 22×22 symmetric matrix

C 22×22 skew-symmetric matrix

D $N \times 22$ rectangular matrix

23 MCQ

Suppose a centered feature matrix $\mathbf{X}$ has shape $N \times 26$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times N$ symmetric matrix

B 26×26 symmetric matrix

C 26×26 skew-symmetric matrix

D $N \times 26$ rectangular matrix

24 MCQ

An observed centered feature matrix $\mathbf{X}$ has shape $N \times 30$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times 30$ rectangular matrix

B $N \times N$ symmetric matrix

C 30×30 symmetric matrix

D 30×30 skew-symmetric matrix

25 MCQ

Consider a centered feature matrix $\mathbf{X}$ has shape $N \times 34$. What is the size and symmetry property of the sample covariance matrix $\mathbf{\Sigma} = \frac{1}{N-1}\mathbf{X}^T\mathbf{X}$?

A $N \times N$ symmetric matrix

B 34×34 skew-symmetric matrix

C $N \times 34$ rectangular matrix

D 34×34 symmetric matrix